

Introduction to Data Science

What is Data Science?

Data Science is an interdisciplinary field focused on extracting knowledge from typically huge datasets. It involves analysis, preparing data for analysis, and presenting findings to inform high-level decisions within an organization. Skills from computer science, mathematics, statistics, information visualization, graphics, and business are incorporated.

Importance and Applications of Data Science

Lots of data is being collected and warehoused, including:

- Web data
- E-commerce data
- Financial transactions
- Bank/credit transactions
- Online trading and purchasing
- Social network data

Examples of the scale of data being processed include:

- Google processing 20 petabytes a day (in 2008)
- Facebook having 60 terabytes of daily logs

The 5 V's of Data Science

Big Data is any data that is expensive to manage and hard to extract value from. The 5 V's help to define big data:

- **Volume**: The size of the data. Big data tools use distributed systems.
- **Velocity**: The latency of data processing relative to the growing demand for interactivity.
 - Example: Social media messages going viral in seconds.
- **Variety**: The diversity of sources, formats, quality, and structures.
 - 80% of world data is unstructured.

Extremely large datasets may be analyzed computationally to reveal patterns, trends, and associations that are not transparent or easy to identify.

Examples of big data in terms of volume include:

- 25+ TBs of log data every day
- 4.6 billion camera phones
- 30 billion RFID tags today
- 12+ TBs of tweet data every day
- 100s of millions of GPS enabled devices sold annually
- 2+ billion people on the Web
- 76 million smart meters in 2009, projected to be 200 million by 2014

Examples of data variety include:

- Relational Data (Tables/Transaction/Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data (Social Network, Semantic Web (RDF))
- Streaming Data (needs to be scanned only once)
- Big Public Data (online, weather, finance, etc.)

Velocity refers to the speed at which data is generated and processed. Late decisions can lead to missed opportunities.

Examples of velocity include:

- E-Promotions: Sending promotions based on a user's current location and purchase history.
- Healthcare monitoring: Sensors monitoring activities and body measurements, requiring immediate reaction to abnormal data.

The ability to collect data is no longer the limiting factor; the ability to manage, analyze, summarize, visualize, and discover knowledge from collected data in a timely and scalable manner is now the challenge.

Real-time analytics and decision requirements include:

- Product Recommendations
- Learning Customer Behavior
- Improving Marketing Effectiveness
- Preventing Fraud

The old model of data generation and consumption involved a few companies generating data and many consuming it. The new model involves everyone generating and consuming data.

Big Data is high-volume, high-velocity, and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision-making, and process automation.

Types of Data

Structured Data

Structured data is data whose elements are addressable for effective analysis. It is organized into a formatted repository, typically a database, and can be stored in SQL databases in tables with rows and columns. It has relational keys and can easily be mapped into pre-designed fields.

Example: Relational data.

Semi-Structured Data

Semi-structured data is information that does not reside in a relational database but has some organizational properties that make it easier to analyze. It eases space and can sometimes be stored in relational databases, though it may be difficult.

Example: XML data.

Unstructured Data

Unstructured data is data that is not organized in a predefined manner or does not have a predefined data model, making it a poor fit for a mainstream relational database. Alternative platforms are used for storing and managing it.

Example: Word, PDF, Text, Media logs.

Challenges and Considerations for Handling Different Data Types

Handling different data types in data science presents unique challenges due to the diverse nature of datasets and the need for analytical rigor. Here are key considerations:

Challenge	Consideration
1. Data Cleaning and Preprocessing	Use imputation for missing values, standardize inconsistent formats, remove or encode outliers appropriately.
2. Mixed Data Types in Datasets	Use appropriate encoding for categorical variables, convert textual data to numerical representations, maintain consistency in numerical data precision.
3. Handling High Cardinality	Use dimensionality reduction techniques, group categories with low occurrences into an "Other" category.
4. Numerical Data Precision	Apply normalization or standardization, be cautious of loss of precision in floating-point operations.
5. Text Data	Preprocess using stemming, lemmatization, and stopword removal, handle multilingual data with language detection and tokenization libraries, use advanced NLP methods.
6. Time Series Data	Convert timestamps to consistent formats, extract features like day, month, year, or lags, resample irregular intervals to a uniform frequency.
7. Image and Multimedia Data	Use pre-trained models for feature extraction, optimize storage using compressed formats.
8. Data Type Mismatches	Use automated tools to identify mismatched types, validate and convert data types explicitly.
9. Missing Data	For numerical data, use mean/median imputation or interpolation. For categorical data, use mode imputation or "Unknown" labels.
10. Large-Scale Data	Use distributed frameworks like Apache Spark or Dask, reduce dimensionality using PCA or feature selection techniques.
11. Bias in Data Types	Apply oversampling (SMOTE) or undersampling to address class imbalance, use balanced accuracy or F1-score for evaluation.
12. Data Visualization	Use specialized plots (scatter plots for numerical, bar plots for categorical), combine data types into summary metrics.

Ethical Considerations in Data Science

Ethical considerations in data science are critical to ensure responsible and fair use of data, algorithms, and outcomes.

Ethical Concern	Examples	Best Practices
1. Data Privacy	Tracking users' online behavior without consent, exposing sensitive information through data breaches.	Implement data anonymization techniques, use encryption for sensitive data storage and transfer.
2. Bias and Fairness	A hiring algorithm discriminates against women, facial recognition systems misclassify certain ethnic groups.	Conduct fairness audits, use diverse and representative datasets, regularly evaluate models for fairness.
3. Informed Consent	Hidden clauses in terms of service agreements, using location data without user consent.	Clearly communicate data collection purposes, obtain explicit consent for data usage, provide opt-out options and transparency.
4. Accountability	No clear ownership of errors in a predictive healthcare model, deploying automated decision systems without oversight.	Assign accountability to specific roles or teams, maintain audit logs, ensure human oversight for critical decisions.
5. Transparency	Complex models deny a loan without explaining why, autonomous vehicles make decisions that cannot be understood.	Use explainable AI techniques, clearly document data sources, provide end-users with interpretable explanations of outcomes.
6. Data Security	Ransomware attacks targeting healthcare datasets, leaks of personally identifiable information (PII).	Employ robust cybersecurity measures, conduct regular security audits, limit access to sensitive data.
7. Misuse of Data	Deepfake generation for spreading misinformation, targeted advertising that manipulates vulnerable populations.	Define ethical use policies for data and models, monitor and restrict usage of harmful tools, collaborate with ethics boards to review applications.
8. Environmental Impact	Training a large NLP model emitting CO2 equivalent to multiple flights.	Optimize algorithms for computational efficiency, use cloud services powered by renewable energy, explore greener machine learning techniques.

9. Intellectual Property	Using copyrighted content for training AI without permission.	Respect copyright and licensing agreements, properly cite data sources and algorithms, develop proprietary datasets ethically.
10. Impact on Jobs and Society	AI replacing customer service roles, predictive policing algorithms targeting marginalized communities.	Engage with stakeholders to assess societal impacts, create transition plans for workers affected by automation, use data science to mitigate inequalities.
11. Dual-Use Dilemma	AI used for medical diagnostics being weaponized in military applications.	Evaluate potential risks during the development phase, restrict access to dual-use technologies.
12. Long-Term Consequences	Data retention policies leading to misuse decades later.	Regularly reassess models and data policies, adopt a precautionary principle when deploying high-impact applications.

Data Science: The Future

The acceleration of data science is due to the enormous volume of data currently available and being generated, combined with the rise of inexpensive computing. Advancing computer memory capacities, enhanced software, competent processors, and numerous data scientists with the skills to use this data make data science essential for the future.

Data Science vs. Big Data

Feature	Data Science	Big Data
Definition	Collection, processing, analyzing, and utilizing data in various operations.	A technique to collect, maintain, and process huge information.
Nature	More conceptual.	A technique for tracking and discovering trends in complex data sets.
Field	A field of study like Computer Science, Applied Statistics, or Applied Math.	A technique for making data more vital and usable by extracting only important information.
Goal	Build data-dominant products for a venture.	Extract important information from huge data within existing traditional aspects.
Tools	SAS, R, Python, etc.	Hadoop, Spark, Flink, etc.

Datafication

Datafication is the process of turning aspects of life into data.

Examples include:

- Twitter datafies stray thoughts.
- LinkedIn datafies professional networks.
- Wal-Mart analyzes buying patterns, stock information, mobile phone location data, social media, and weather information to send targeted vouchers.

We are being datafied intentionally when we like something online, unintentionally through cookies, and passively via sensors and cameras.

The Data Science Lifecycle

Stage	Description
1. Data Acquisition	Gathering raw structured and unstructured data through data entry, signal reception, and data extraction.
2. Data Processing	Taking the raw data and putting it in a form that can be used through data mining, data warehousing, data cleansing, data capture, and data architecture.
3. Data Analysis	Examining patterns, ranges, and biases to determine how useful the prepared data will be in predictive analysis through exploratory, predictive, and qualitative analysis.
4. Data Communication	Preparing analyses in easily readable forms such as charts, graphs, and reports for data reporting, data visualization, and business intelligence.

A Data Science Profile

On a daily basis, a data scientist may:

1. Discover patterns and trends in datasets to get insights.
2. Create forecasting algorithms and data models.
3. Improve the quality of data or product offerings by utilizing machine learning techniques.
4. Distribute suggestions to other teams and top management.
5. Use data tools such as R, SAS, Python, or SQL in data analysis.
6. Top the field of data science innovations.

Data Science Job Market in India

The Indian analytics industry is predicted to escalate to USD 98 billion in 2025 and nearly USD 119 billion in 2026. The demand for data scientists is at an all-time high in India, with analysts predicting around 11 million job openings in data science by 2026 in India alone.

Top Data Scientist Recruiters

- Amazon
- IBM
- Deloitte
- Accenture
- Fractal Analytics
- Citrix
- LinkedIn
- Myntra
- MuSigma
- Dexlock
- Flipkart
- Rudder Analytics

Impact of Applying Data Science in Business Scenario

As a business, the goal is to provide better products to consumers so that they return. Data science helps in developing user-centric products by:

- Analyzing customer reviews
- Analyzing current market trends
- Comparing two products
- Finalizing the best products with the ability to attract customers and hold on to them for a longer time

Specific Impacts:

1. **Reduces Inefficiencies:** Data scientists track company-wide metrics to pinpoint areas for improvement, lowering overall costs and boosting return-on-investment.
2. **Predicts Trends and Customer Behavior:** Data-based predictions determine peak customer shopping times, adjust staff levels, identify early buyer trends, and implement appropriate promotional campaigns.
3. **Enables Competitor Research:** Effective competitor research helps businesses make competitive pricing decisions, reach new markets, and stay up to date with changes in consumer behavior.
4. **Allows Testing of Business Initiatives:** Consistent, long-term testing enables companies to drive incremental revenue gains through successful marketing campaigns, product launches, employee satisfaction, and website optimization.
5. **Develops Market Understanding:** Data science enables businesses to reshape their products and services to fit with a shifting marketplace by mining information from various sources.
6. **Informs Hiring Decisions:** Data science bridges the gap between prospects that look good on paper and perform well in practice by using evidence to improve hiring practices.

Data Science Use Cases

Case Study: Uber

Intelligent Ride Management

Every ride booked gives Uber information about the user, from preferred pick-up points to most frequent destinations. Uber records behavior, preferences, interests, and even phone battery levels to determine demand, allocate resources, and set fares to maximize profit. Data is collected smartly and cleaned to find relevance, including data from drivers, irrespective of whether they are carrying passengers or not. Drivers' speed and acceleration, their location, and data on whether they work for a competitor are all retrievable. The AI team then draws inferences from understanding traffic patterns, ETA, journey times, and surge pricing.

Pricing

Longer or shorter journey times give Uber the advantage of real-time pricing. Data Scientists use predictive modelling in real-time to estimate demand and execute informed pricing. Supply and demand analysis algorithms monitor traffic conditions and journey duration in real-time, adjusting pricing as demand changes.

Mapping

Data scientists build models using several map services like Google Maps. Data from millions of trips enabled Uber to offer pick/drop services to specific gates.

Driver-Less Cars

The Uber AI Labs at Pittsburgh is running experiments with autonomous cars. By improving their mapping system with deep learning, machines learn better with more exposure. As more trips are taken and more models are built using AI, the machines are getting smarter about handling unprecedented scenarios.